

- 6** The lines l_1 and l_2 have equations $\mathbf{r} = 2\mathbf{i} + \mathbf{k} + \lambda(\mathbf{i} - \mathbf{j} + 2\mathbf{k})$ and $\mathbf{r} = 2\mathbf{j} + 6\mathbf{k} + \mu(\mathbf{i} + 2\mathbf{j} - 2\mathbf{k})$ respectively.

The point P on l_1 and the point Q on l_2 are such that PQ is perpendicular to both l_1 and l_2 .

- (a) Find the length PQ . [5]

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

The plane Π_1 contains PQ and l_1 .

The plane Π_2 contains PQ and l_2 .

- (b) (i)** Write down an equation of Π_1 , giving your answer in the form $\mathbf{r} = \mathbf{a} + s\mathbf{b} + t\mathbf{c}$. [1]

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- (ii) Find an equation of Π_2 , giving your answer in the form $ax + by + cz = d$. [4]

[illegible]

- (c) Find the acute angle between Π_1 and Π_2 . [5]

[illegible]